<u>ICWaR Course</u>

WR 203

Title: Applied geochemical modeling and Water quality analysis (2:1)

Instructor: Prof. Praveen C Ramamurthy

Term: August - December

Contents:

Simulation and modelling

- Introduction to the Storm Water Management Model (SWMM) in Urban Catchments (3 hr)
- Tutorial on Remote Sensing (3 hr)
- Tutorial on BRAT (Basic Radar Altimetry Toolbox) (3 hr)
- Introduction to Climate/Earth System Modeling (3 hr)
- Modelling and Simulations of Liquids for Chemical and Thermophysical Properties (3 hr)
- Mountain water resources (1.5 hr)
- Water budget equation and hydrological cycle (1.5 hr)
- Water Sampling and applications (2 hr)

Water Quality (10 hr)

- To assess the alkalinity of a given water sample. (1 hr)
- Determine chloride ion concentration in a water sample. (1 hr)
- To assess the total solids of a given sample of water. (1 hr)
- Measure Total hardness using dye indicators. (1 hr)
- To introduce concepts of total coliforms using the multiple-tube fermentation technique. (2 hr)
- To assess the color of the given water sample. (1 hr)
- Determine the DO content of a given sample. (1 hr)
- To assess the pH value of the given water samples. (1 hr)
- To assess the residual chlorine of the given water sample. (1 hr)

Basic electronic instrumentation (5 hr)

Measurements of ionic contaminants using an ICPMS and IC (5 hr)